The Bottom of the Shub–Smale Tau Conjecture: an Exact Census of Integer Roots for Constant-Free Straight-Line Programs of Length at Most Eight

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Abstract

For a univariate integer polynomial f, let $\tau(f)$ be the minimum number of $+, -, \times$ gates needed to compute f from x and the constants $-1, 0, 1$, and let $z(f)$ be its number of distinct integer roots. The Shub–Smale tau conjecture (Smale's fourth problem) asserts $z(f) \le (1+\tau(f))^\kappa$ for a universal κ; it implies $\mathrm{P}_{\mathbb{C}} \ne \mathrm{NP}_{\mathbb{C}}$ in the Blum–Shub–Smale model and $\mathrm{VP}^0 \ne \mathrm{VNP}^0$, and it is open even for $\kappa = 1$. We report the first exact census of the conjecture's growth function at the bottom of its ladder: writing $z_{\max}(\tau)$ for the maximum of $z(f)$ over all nonzero f with $\tau(f) \le \tau$, we prove by exhaustive, exactly verified computation that

$$z_{\max}(1), \ldots, z_{\max}(8) \;=\; 1,\, 2,\, 3,\, 3,\, 4,\, 5,\, 5,\, 6.$$

In particular the minimum cost of 4 distinct integer roots is 5 gates, of 5 roots is 6 gates, and (new in this version) of 6 roots is exactly 8 gates, by an explicit verified witness: $q(q-2)(q-6)$ for $q = x^2 - x$, computable in 8 gates by chained subtraction sharing; the growth function has plateaus at $\tau = 4$ and $\tau = 7$, so an extra gate does not always buy an extra root, and (new in this version) $z_{\max}(8) = 6$ exactly: no 8-gate program of any shape has 7 distinct integer roots, so the seven-root threshold lies in $\{9, 10\}$. The censuses are decision-complete: depth 6 required the exact construction of all 25,844,905 reachable computation states; depth 7 a complete scan of 2,013,706 new polynomials, using a "last-gate" lemma that decides one depth beyond any exhausted frontier without storing the next frontier; and depth 8 (new in this version) the first complete construction of the depth-7 frontier itself: 1,048,460,912 states, built out-of-core by hash-partitioned external deduplication and scanned exhaustively ($\sim 2 \times 10^{11}$ last-gate applications) by a validated multiprocess engine. The enumerator is anchored on Markström's published exhaustive census of integer targets (all fourteen anchor values reproduced exactly) and cross-checked against an independent computer-algebra root counter on 284 polynomials. The six-root threshold was closed by a case split: the last gate of an 8-gate 6-rooter must involve the seventh value, and if it is a multiplication its root count is a union of two known root sets, so that case reduces to a co-occurrence query over the depth-6 frontier; the scan found 408 witnesses. We further prove two elementary stall theorems explaining the observed record mechanisms: for any monic $h \in \mathbb{Z}[x]$ of degree at least 2, towers built by iterating h alone have depth-independent integer root counts (for $h = x^2 - 2$, the iterates $h^{\circ k}(x) - x$ keep exactly 2 integer roots against 2^k real roots), and across the quadratic family $h_c = x^2 - c$ with $c \le 200$ the maximum tower yield is 5, attained only at $c = 2$; a second yield series at $c = m^2+m+1$, produced by genuine integer 2-cycles, is identified and closed by the classical fact that integer polynomial cycles have length at most 2. All computations are exact integer arithmetic, reproducible from committed code with hypotheses declared before each run; four of our own pre-registered mathematical predictions (and two solver-tractability judgments) were refuted by the machine and are reported as such. We also report the exact ladders of the digit-restricted census (the form of the conjecture Rojas proved sufficient): the maximum number of ODD roots grows as $1, 2, 2, 2, 2, 3, 4$ through $\tau = 7$, with its own extremal family $(x^2 - 1)(x^2 - 9)$, distinct from the unrestricted records. Finally we place the census against its two neighbouring worlds, measured on the same enumerated programs. Over $\mathbb{F}_p$ the analogous maximum is *exactly* $2^{\tau-1}$ for every τ (proved here; the extremal program is k squarings and one subtraction, and the Fermat primes that the small cases suggest are not required, only a prime $p \equiv 1 \bmod 2^k$). Over $\mathbb{R}$ it is $1, 2, 3, 4, 6, 8$ through $\tau = 6$, exhaustively, by exact Sturm counting with no floating point; its records are generated by a step $g \mapsto g \cdot (g - x^2)$ that costs two gates and doubles the real root count, beating a Chebyshev tower's three gates per doubling, and that itself stalls after three iterations. The three ladders agree at $\tau \le 2$ and separate at $\tau = 3$ and $\tau = 4$ respectively; over $\mathbb{F}_p$ the degree ceiling $2^{\tau-1}$ is saturated at every τ, over $\mathbb{R}$ it is not, and over $\mathbb{Z}$ the census stays linear. A five-gate program with six distinct *real* roots and only two integer ones exhibits the difference concretely. This is an experimental-mathematics record in the tradition of Markström's census: it decides nothing about the conjecture asymptotically, and says so.

1 Introduction

1.1 The conjecture

A *constant-free straight-line program* (SLP) over $\mathbb{Z}[x]$ is a sequence $(1, x, f_2, \ldots, f_N)$ in which every f_i ($i \ge 2$) is a sum, difference, or product of two earlier entries; following [1, 8] we write $\tau(f)$ for the least $N - 1$ over all such programs with $f_N = f$. Equivalently [6], $\tau(f)$ is the minimal number of operation gates of an arithmetic circuit over the inputs x and the free constants $-1, 0, 1$. Let $z(f)$ denote the number of distinct roots of f in $\mathbb{Z}$.

Conjecture 1.1 (Shub–Smale [1]; Smale's fourth problem [2])**.** *There is a universal constant $\kappa \ge 1$ such that every nonzero $f \in \mathbb{Z}[x]$ satisfies $z(f) \le (1 + \tau(f))^\kappa$.*

The conjecture fails for $\kappa < 1$: the geometric-progression products $(x-2)(x-4)\cdots(x-2^{2^j})$ acquire one root per $O(1)$ gates [8]. It is open even for $\kappa = 1$. Its significance is structural: Shub and Smale proved that it implies $\mathrm{P}_{\mathbb{C}} \ne \mathrm{NP}_{\mathbb{C}}$ in the Blum–Shub–Smale model [1, 3], the proof reducing to the hardness of computing all sequences $(m_n \cdot n!)$; Bürgisser [5], building on Koiran [4], proved that it implies $\mathrm{VP}^0 \ne \mathrm{VNP}^0$, and that the single sequence $(n!)$ being hard to compute already yields that separation. Easy factorials would give nonuniform polynomial-time integer factoring [3, 9]; with integer division allowed, $n!$ *is* easy [11], so the restriction to $+, -, \times$ carries the entire content. The real-zeros analogue of the conjecture is false: iterating the logistic map produces 2^j real roots at cost $O(j)$ [8], which motivated Koiran's real tau conjecture for structured expressions [13, 14, 15, 16, 17]. On the lower-bound side, no nontrivial lower bound on $\tau(n!)$ is known [4]; on the reduction side, Rojas proved that a p-adic digit version of the conjecture, for any fixed prime, already implies the full statement, via a p-independent subquadratic bound on the valuation spectrum of roots [8]. Recent work applies the conjecture to explicit exponential lower bounds [18] and tightens the bridge from BSS separations to constant-free Valiant separations [19]; the current survey reference is [6, §4.6].

1.2 What this paper does

The literature contains, to our knowledge, exactly one exhaustive computation in this model: Markström's census of *integer targets* (the straight-line complexity of small factorials and primorials), complete to program length 11 on 2013-era hardware [7]. No published computation addresses the *polynomial* side, i.e. the actual growth function of the conjecture. We define

$$z_{\max}(\tau) \;=\; \max\{\, z(f) \;:\; f \in \mathbb{Z}[x] \setminus \{0\},\; \tau(f) \leq \tau \,\}$$

and determine it exactly for $\tau \leq 7$, together with the full record galleries, explicit optimal witnesses, and the exact cost thresholds for $3, 4, 5$ and (within one gate) 6 distinct integer roots. We also prove two elementary theorems that explain the observed record mechanisms and close entire construction classes (Section 6), and we measure the quadratic tower family exhaustively (Section 7).

Our contributions:

1. **The census** (Section 4): $z_{\max}(1..7) = 1, 2, 3, 3, 4, 5, 5$, decision-complete, with per-depth state and polynomial counts, record galleries and replayed witness programs. Plateaus occur at $\tau = 4$ and $\tau = 7$.

2. **Exact thresholds**: the minimum τ admitting 3 distinct integer roots is 3 ($x^3 - x$); for 4 roots it is 5; for 5 roots it is 6; for 6 roots it is exactly 8 (Section 4.1); for 7 roots it is 9 or 10 (Section 4.2: the complete depth-8 census excludes 8; a 10-gate witness exists).

3. **A method lemma** (Lemma 2.5): every polynomial of cost $d + 1$ is one gate over a normalized depth-d state, so $z_{\max}(d+1)$ is computable from an exhausted depth-d frontier without storing the depth-$(d + 1)$ frontier. This bought depth 7 (a $\sim 10^9$-state frontier avoided).

4. **Stall theorems** (Section 6): iterating any fixed monic map of degree ≥ 2 yields depth-independent integer root counts; for the Chebyshev-conjugate map $x^2 - 2$, whose difference-of-squares splittings generate all census records at depths 5–7, the exact stable core is $\{0, \pm 1, \pm 2\}$.

5. **The family measurement** (Section 7): across $h_c = x^2 - c$, $c \leq 200$, and all tower shapes to four iterations, the maximum integer-root yield is 5, attained only at $c = 2$; two yield-4 series exist, $c = m(m + 1)$ (fixed and anti-fixed points) and $c = m^2 + m + 1$ (genuine 2-cycles), and the classical bound on integer polynomial cycle lengths shows this inventory is complete.

6. **The seven-root threshold** (Section 8): the upper bound is verified by two structurally different 10-gate programs, executed gate by gate, one folding ($q(q - 2)(q - 6) \cdot (x - 4)$, root set the interval $\{-2, \ldots, 4\}$) and one squaring ($x^2(x^2 - 1)(x^2 - 4)(x^2 - 16)$, root set the tower $\{0, \pm 1, \pm 2, \pm 4\}$); the squaring shape generalizes to a family giving $2m + 3$ roots in $3m + 4$ gates. On the lower side, no 9-gate program whose final gate is a *multiplication*

reaches 7 roots, over all 1,048,460,912 depth-7 states (maximum union size exactly 6). We also prove a confinement lemma turning the additive case's evaluation window into a decidable criterion: if f has 7 distinct integer roots and trailing coefficient c with $|c| \leq 395$, every root lies in $[-32, 32]$; the constant is sharp.

7. **The three worlds** (Section 9): on the same enumerated programs, $z^{p}_{\max}(\tau) = 2^{\tau-1}$ exactly for every τ [D], while $z^{\mathbb{R}}_{\max}(1..6) = 1, 2, 3, 4, 6, 8$ [MV] and the integer census stays linear. What limits $\mathbb{Z}$ is not degree and not splitting, both of which cheap programs reach at exponential scale, but *distinctness*: five gates already give a completely split polynomial of degree 16 over $\mathbb{Z}$ with two distinct roots.

8. **Anchoring and audit** (Section 3): the enumerator reproduces all fourteen of Markström's published anchor values; an independent computer-algebra cross-check agrees on 284 polynomials; every experiment's hypothesis, budget and kill criterion were committed before its run, and two of our pre-registered predictions were refuted by the machine (both disclosed below).

Every claim in this paper is labeled: **[MV]** machine-verified in exact arithmetic from committed code, **[D]** proved here (elementary, self-contained), or **[C]** conjectural/observational. Nothing here decides the conjecture in either direction, and no asymptotic consequence is claimed: the census maps where the conjecture lives at the bottom of its ladder, in the same genre as [7].

2 Model, normalization, and the census method

Definition 2.1. Programs operate on values in $\mathbb{Z}[x]$. The inputs are -1, 1 and x; each gate applies $+$, $-$ or $\times$ to two (not necessarily distinct) earlier values; the length of a program is its number of gates; $\tau(f)$ is the minimal length of a program computing f at some gate.

Lemma 2.2 (free 0 is redundant)**.** *Adding* 0 *as a fourth free input does not change* τ*.* [D]

Proof. In any program using the input 0: $a + 0$ and $a - 0$ duplicate a; $a \times 0$ computes 0; $0 - a$ equals $(-1) \times a$, one gate either way. Replacing each use and deleting duplicated gates never lengthens the program. □

Lemma 2.3 (normalization)**.** *Every* f *has an optimal program in which no gate recomputes an input or an earlier value, and no gate computes* 0*.* [D]

Proof. A gate duplicating an available value can be deleted, rewiring its uses to the original; a computed 0 is usable only in the redundant patterns of Lemma 2.2. Each deletion shortens the program, so an optimal one contains neither. □

Lemma 2.4 (reached-set sufficiency)**.** *The extensions available to a program depend only on the* set *of values it has computed. Consequently a breadth-first search over reached sets ("states"), with set-level deduplication, visits every normalized reachable state of each depth exactly once.* [D]

This is the polynomial analogue of the range-isomorphism reduction of [7]. The census algorithm is exactly this BFS, in exact integer arithmetic on dense coefficient tuples, with each new polynomial's first appearance depth recorded (that depth is $\tau(f)$, by Lemmas 2.3 and 2.4). Integer roots are counted exactly: strip the power of x, then test the divisors of the trailing coefficient (both signs).

Lemma 2.5 (last gate)**.** *Let $\mathcal{F}_d$ be the set of normalized reached states of depth d. Every f with $\tau(f) = d + 1$ is $a \circ b$ for a single gate $\circ$ and operands a, b available in some state of $\mathcal{F}_d$. Hence $z_{\max}(d + 1) = \max\big(z_{\max}(d),\ \max z(a \circ b)\big)$ over one op over every state of $\mathcal{F}_d$, and this maximum is computable without constructing $\mathcal{F}_{d+1}$.* [D]

Proof. Take an optimal program of length $d + 1$ for f and normalize it (Lemma 2.3; deletions would shorten it, contradicting optimality, so it is already normalized). Its first d gates form a normalized reached state of depth d, and its last gate is one op over that state's available values. □

3 Anchoring and audit discipline

Markström anchor [MV]. Restricted to the integer model (start value 1, positive normalized values), our enumerator reproduces Figure 1 of [7] exactly at every compared depth: reached-set sizes $2, 4, 9, 26, 102, 562, 4363$ and complete initial intervals $2, 4, 6, 12, 40, 112, 310$ for lengths 1 through 7: fourteen of fourteen anchor values.

Independent cross-check [MV]. An independent root counter (SymPy's exact factorization) agrees with ours on all 284 polynomials checked: every polynomial of $\tau \leq 3$ and every record polynomial of the censuses.

Engine re-derivation [MV]. The depth-7 computation uses a reimplemented, memoized engine; before its new output was trusted, it was required to re-derive every prior census value (state counts 9/98/1462/29506/778087, new-polynomial counts through depth 6, and the record sets), and did.

Pre-registration. Every computation ran under a hypothesis file committed to the repository *before* the run, with falsifiable predictions, an explicit statement of what a pass and a fail would prove, a compute budget and a kill criterion. Three committed predictions were refuted by the machine: we predicted $z_{\max}(6) = 4$ (the census returned 5; Section 4); we predicted zero tower yield off the fixed-point series (a second series exists; Section 7); and we predicted the six-root multiplication case empty at 8 gates (the scan found 408 witnesses; Section 4.1). Each refutation is reported alongside what it taught. One tooling incident is likewise on record: divisor-based root counting is infeasible against tower constant terms of size c^{2^k}, and was replaced, for the family measurement only, by direct evaluation over a window proved correct in advance (Section 7), cross-checked against the divisor method where both run.

4 The census

Theorem 4.1 (the census; MV)**.** *In the model of Definition 2.1,*

τ	1	2	3	4	5	6	7	8
$z_{\max}(\tau)$	1	2	3	3	4	5	5	6

and the censuses behind each value are decision-complete.

The supporting exhaustive counts are:

τ	1	2	3	4	5	6	7	8
reached states	9	98	1,462	29,506	778,087	25,844,905	1,048,460,912	(scan)
new polynomials	9	34	177	1,249	11,377	134,494	2,013,706	(see text)
$z_{\max}$	1	2	3	3	4	5	5	6

Depth 7 was decided by Lemma 2.5 over the exhausted depth-6 frontier; among its 2,013,706 new polynomials the distinct-integer-root histogram is 742,635 rootless, 904,421 with one, 299,488 with two, 63,903 with three, 3,196 with four, 63 with five, and none with six or more.

Theorem 4.2 (cost thresholds; MV)**.** *The minimum τ admitting k distinct integer roots is:* $k = 1\colon 1$*;* $k = 2\colon 2$*;* $k = 3\colon 3$*;* $k = 4\colon 5$*;* $k = 5\colon 6$*;* $k = 6\colon 8$*; and for* $k = 7$ *it lies in* $\{9, 10\}$.

The $k = 6$ lower bound is Theorem 4.1 ($z_{\max}(7) = 5$); the upper bound is the explicit 8-gate witness of Section 4.1. Representative optimal witnesses, replayed gate by gate in exact arithmetic from the committed artifacts:

τ	record	witness program
3	$x^3 - x$, roots $\{0, \pm 1\}$	$x{\cdot}x;\ x^2{-}1;\ x{\cdot}(x^2{-}1)$
4	$-x(x{+}1)(x{+}2)$, roots $\{0, -1, -2\}$	$-1{-}x;\ 1{-}({-}1{-}x);\ x({-}1{-}x);\ (x^2{+}x)\ldots$
5	$x^2 - (x^2{-}2)^2 = -(x^2{-}1)(x^2{-}4)$	$-2;\ x^2;\ x^2{-}2;\ (x^2{-}2)^2;\ x^2 - (x^2{-}2)^2$
6	$\mp x(x^2{-}1)(x^2{-}4)$, roots $\{0, \pm 1, \pm 2\}$	the $\tau{=}5$ record, then $\times\, x$

Plateaus. $z_{\max}$ pauses at $\tau = 4$ and $\tau = 7$. The record galleries explain the asymmetry: the step $5 \to 6$ succeeds because multiplying by the input x costs one gate and adjoins the root 0 to a record avoiding it; the steps $3 \to 4$ and $6 \to 7$ fail because every candidate next root beyond the "free" set $\{0, \pm 1, \pm 2\}$ requires a built constant, and constants cost gates. The $\tau = 5$ records (ten polynomials) and $\tau = 6$ records (four) are all difference-of-squares splittings over the map $x^2 - 2$ (Section 6); at $\tau = 7$ the sixty-three five-rooters appear, but no six-rooter.

4.1 Closing the six-root window

Version 0.01 of this paper left the six-root threshold in $[8, 9]$. It is 8:

Theorem 4.3 (MV, both directions)**.** *There is an* 8*-gate constant-free SLP whose output has* 6 *distinct integer roots, and none with* 7 *gates. An explicit witness (replayed gate by gate in exact arithmetic) is, with* $q = x(x-1)$*:*

$$x{-}1; \quad 2; \quad q = x \cdot (x{-}1); \quad 4; \quad q{-}2; \quad (q{-}2){-}4 = q{-}6; \quad q(q{-}2); \quad f = q(q{-}2) \cdot (q{-}6),$$

whose root set is $\{-2, -1, 0, 1, 2, 3\}$*. Hence* $z_{\max}(8) \geq 6$.

The method is a case split. By Lemma 2.5 at depth 8, an 8-gate 6-rooter is one gate over a depth-7 state $S \cup \{v\}$; since $z_{\max}(7) = 5$, the final gate must involve the seventh value v. If that gate is a multiplication, $f = v \cdot b$ and $z(f) = |R_v \cup R_b|$: a pure co-occurrence condition on known root sets, decided exactly by one scan of the 25,844,905 depth-6 states (2h23m single-core): 408 hits, three reconstructed as explicit programs and verified independently of the scan machinery. Two committed expectations fell in the process: we had predicted this case empty (our schema hunt costed the witness family at 9 gates because it built the constants $\{2, 6\}$ independently; the scan's witness builds $\{2, 4\}$ and reuses the subtraction chain $q \mapsto q - 2 \mapsto q - 6$: a one-gate saving our cost model missed), and we had predicted that all $\tau \leq 7$ five-rooters have root set $\{0, \pm 1, \pm 2\}$: in fact the 67 five-rooters realize seven root-set patterns, including the non-consecutive $\{-1, 0, 1, 2, 4\}$ and $\{-4, -2, -1, 0, 1\}$, and it is precisely such shifted and punctured sets that leave room for a co-occurring factor to complete six roots.

4.2 The depth-eight resolution

Version 0.03 completes depth 8 in full:

Theorem 4.4 (MV)**.** *$z_{\max}(8) = 6$, and no 8-gate program of ANY final-gate shape computes a polynomial with 7 distinct integer roots. Hence the seven-root threshold is 9 or 10.*

The lower side ($z_{\max}(8) \geq 6$) is Theorem 4.3. The upper side required the whole depth-7 frontier: it was CONSTRUCTED, for the first time, as 1,048,460,912 reached-set states (out-of-core: 2.43 billion raw 28-byte successor rows, hash-partitioned into 256 files that deduplicate independently in memory; 2h50m), with an exact internal cross-anchor: the build's count of new depth-7 polynomials, 2,013,706, equals the value computed independently by the depth-7 scan of Section 4. The frontier was then scanned exhaustively by the last-gate lemma ($\sim 2 \times 10^{11}$ gate applications, 20 processes, 6h48m): zero results with 7 or more distinct integer roots. Every pipeline stage passed a known-answer gate first (the identical machinery one level down reproduces the depth-6 frontier and the $z_{\max}(7) = 5$ census exactly). We record for methods completeness that two general-purpose solver engines (nonlinear integer arithmetic and pure bit-vector SAT over a gate-evaluation encoding) were validated semantically on pinned witnesses yet could not complete even the smallest free-structure known-answer search within generous budgets: at this problem's scale, exhaustive construction beat search.

4.3 The digit-restricted census

Rojas' reduction makes a digit-restricted bound SUFFICIENT for the full conjecture: bounding only roots $\equiv 1 \pmod p$ for one fixed prime suffices. We measured that restricted growth function exactly through $\tau = 7$: the maximum number of odd roots is $1, 2, 2, 2, 2, 3, 4$, and for roots $\equiv 1 \pmod 3$: $1, 1, 1, 2, 2, 3, 3$. Our committed prediction (odd maximum 3) was refuted: the digit-restricted world has its OWN extremal family, e.g. $(x^2-1)(x^2-9)$ at exactly 7 gates with all four roots odd: symmetric products over odd squares, distinct from the unrestricted records (which crowd $\{0, \pm 1, \pm 2\}$ and are digit-poor). The sufficient form of the conjecture is thus measurably harder to grow AND mechanically independent at the bottom of the ladder.

5 Record mechanisms

The census records decompose into a small move inventory [MV for the instances; observational as a classification]: linear factors $x - a$ for available constants a; block translation at one gate per unit shift; multiplication by the input x (one gate, adjoins 0); and difference-of-squares splittings $B^2 - A^2 = (B - A)(B + A)$ at two gates given A and B. The depth-5 record $x^2 - (x^2 - 2)^2 = -(x-1)(x+1)(x-2)(x+2)$ is the fundamental instance: its inner map $C(x) = x^2 - 2$ is the Chebyshev-conjugate doubling map ($h(z) = z + 1/z$ satisfies $h(z)^2 - 2 = h(z^2)$), i.e. the record is the integer shadow of exactly the iteration that makes the *real* analogue of the conjecture false. Over $\mathbb{Z}$, that factory is sterile beyond one splitting, which the next section proves.

6 Stall theorems

Lemma 6.1 (escape)**.** *Let $h = x^d + \sum_{i<d} a_i x^i \in \mathbb{Z}[x]$, $d \geq 2$, and $R = 1 + \sum_{i<d} |a_i|$. If $|x| \geq R + 1$ then $|h(x)| \geq |x| + 1$.* [D]

Proof. $\sum_{i<d} |a_i|\,|x|^i \le (R-1)|x|^{d-1}$ for $|x| \ge 1$, so $|h(x)| \ge |x|^{d-1}(|x| - (R-1)) \ge 2|x|^{d-1} \ge 2|x| \ge |x| + 1$ for $|x| \ge R+1$. □

Theorem 6.2 (monic stall; D)**.** *Fix a monic $h \in \mathbb{Z}[x]$ of degree ≥ 2. There is a constant $Z(h)$ such that for every $k \ge 1$, both $h^{\circ k}(x) - x$ and the difference-of-squares towers $G_k = h^{\circ(k-1)}(x)^2 - h^{\circ k}(x)^2$ have at most $Z(h)$ distinct integer roots. One may take $Z(h) = \#([-M, M] \cap \mathbb{Z})$ where $M = \max(R+1, \max|a|)$ over the integer solutions a of $h(y) = \pm y$.*

Proof. Integer roots of $h^{\circ k}(x) - x$ are periodic points of h; by Lemma 6.1 all integer periodic points lie in $[-R-1, R+1]$. For G_k, factor $G_k = (h^{\circ(k-1)} - h^{\circ k})(h^{\circ(k-1)} + h^{\circ k})$: an integer root x has $y = h^{\circ(k-1)}(x)$ solving $h(y) = \pm y$, so x lies in the increasing union $\bigcup_j (h^{\circ j})^{-1}(A) \cap \mathbb{Z}$ for the finite set A of such y. If $|x| > M$ then $|h^{\circ j}(x)| \ge |x| + j > \max_{a \in A} |a|$ for all j by Lemma 6.1 and induction, so no element of the union exceeds M in absolute value; the union is contained in a fixed finite set and stabilizes. □

Proposition 6.3 (the Chebyshev core; D, MV spot-checks)**.** *For $C(x) = x^2 - 2$: the integer periodic points are exactly $\{2, -1\}$; for every $k \ge 1$ the polynomial $C^{\circ k}(x) - x$ has exactly 2 distinct integer roots while having 2^k distinct real roots and $\tau \le 2k+2$; and the towers G_k have root set $\{\pm1, \pm2\}$ for $k = 1$ and exactly $\{0, \pm1, \pm2\}$ for every $k \ge 2$.*

Proof. Orbits: $0 \mapsto -2 \mapsto 2 \mapsto 2$ and $1 \mapsto -1 \mapsto -1$; by Lemma 6.1 (with $R = 3$; direct check for $|x| \ge 3$) all other integer orbits escape, so the only cycles are the fixed points 2 and -1, giving the first two claims (the real-root count is the classical Chebyshev conjugacy). For G_k: $h(y) = y$ has integer solutions $\{2, -1\}$ and $h(y) = -y$ has $\{-2, 1\}$; integer preimages are $C^{-1}(2) = \{\pm2\}$, $C^{-1}(-1) = \{\pm1\}$, $C^{-1}(-2) = \{0\}$, $C^{-1}(1) = C^{-1}(0) = \emptyset$, so the preimage closure of $\{\pm1, \pm2\}$ is $\{0, \pm1, \pm2\}$ and is stable. □

Theorem 6.2 closes a construction class: no family built by iterating one fixed monic map can witness superpolynomial z versus τ; a refutation of the conjecture, if one exists, must build fresh constants or mix maps, paying gates for them. That is precisely the regime the census measures.

7 The quadratic family

We measured all tower shapes ($h^{\circ k}(x) - x$ and $h^{\circ i}(x)^2 - h^{\circ j}(x)^2$, $0 \le i < j \le 4$) across $h_c = x^2 - c$ for $1 \le c \le 200$, in exact arithmetic. Root finding here cannot use trailing-coefficient divisors (those constants reach $c^{2^k} \sim 10^{18}$); instead, by Lemma 6.1, every integer root of such a shape lies in $[-(c+1), c+1]$, and direct evaluation over that window is exact. The two finders agree on every shape where both are feasible ($c \le 10$, $j \le 2$) [MV].

Theorem 7.1 (family census; MV)**.** *For $1 \le c \le 200$ and the shapes above: the maximum number of distinct integer roots is 5, attained only at $c = 2$ (the set $\{0, \pm1, \pm2\}$ of Proposition 6.3). Nonzero yields occur exactly on two series: $c = m(m+1)$, yield 4 for $m \ge 2$ with root set $\{\pm m, \pm(m+1)\}$ (from the fixed points $\{m+1, -m\}$ and anti-fixed points $\{m, -m-1\}$), and $c = m^2 + m + 1$, yield 4 with the same root sets, produced by the genuine 2-cycles $m \mapsto -m-1 \mapsto m$ (for $c = 1$: yield 3, root set $\{0, \pm1\}$, from the cycle $0 \leftrightarrow -1$).*

The second series refuted our committed prediction that only the fixed-point series yields roots. It is, however, the *last* such surprise available to this construction class:

Proposition 7.2 (cycle ceiling; D, classical)**.** *For any $f \in \mathbb{Z}[x]$, every cycle of f in $\mathbb{Z}$ has length at most 2. Consequently fixed points, anti-fixed points and 2-cycles are the complete periodic inventory harvestable by tower shapes over any integer polynomial map.*

Proof. For integers $a \neq b$, $(a-b) \mid (f(a)-f(b))$. Around a cycle $a_0 \mapsto a_1 \mapsto \cdots \mapsto a_r = a_0$ of distinct points, the differences $d_i = a_{i+1} - a_i$ satisfy $d_i \mid d_{i+1}$ cyclically, so all $|d_i|$ equal one value $d > 0$, and the cycle is a closed walk on the arithmetic line $a_0 + d\,\mathbb{Z}$ with steps $\pm d$. If two consecutive steps have opposite signs then $a_{i+2} = a_i$, which forces $r = 2$; if all steps have the same sign the walk never closes. Hence $r \leq 2$. The fact is classical in the polynomial-cycles literature (see Narkiewicz [21]); the proof is included to keep the paper self-contained. □

Yield per gate across the family is maximized at $c = 2$ and decreases in c (the yield is bounded by 5 while building c costs $\tau(c) \sim \log c$ gates, measured exactly against our anchored integer census): within this family, the "parameterized loophole" left open by Theorem 6.2 is empty [MV for $c \leq 200$; C beyond].

8 The seven-root threshold

Theorem 4.4 left the threshold for 7 distinct integer roots at 9 or 10. This section closes the upper side constructively, decides the multiplicative half of the ninth gate exhaustively, and states exactly what remains conditional on the additive half.

8.1 Two ten-gate witnesses

Version 0.03 asserted a 10-gate witness in passing. We now give two, written as explicit programs and executed gate by gate in the same exact arithmetic as the census; they are structurally different, and the difference is the point.

Proposition 8.1 (upper bound; MV)**.** *Each of the following constant-free programs has length* 10 *and computes a polynomial with* 7 *distinct integer roots.*

(A) *With $q = x^2 - x$: $p_A = q(q-2)(q-6) \cdot (x-4)$, root set the interval $\{-2, \ldots, 4\}$, height* 56*.*

(B) $p_B = x^2(x^2-1)(x^2-4)(x^2-16)$*, root set the tower $\{0, \pm 1, \pm 2, \pm 4\}$, height* 84*.*

Hence the seven-root threshold is at most 10*.*

Both gate counts are exact against the model's conventions because the programs are run as data rather than counted by hand; neither is claimed to be optimal, which is the business of the scan below. Witness (A) costs nothing extra for its third factor, since $q - 6 = (q-2) - 4$ and the constant 4 is already in the state of the exhaustive 8-gate six-rooter $q(q-2)(q-6)$ of Theorem 4.3; appending $(x-4)$ costs one subtraction and one multiplication. Witness (B) is cheap for a different reason: each of its constants is a single squaring past the previous one ($2 \to 4 \to 16$), so it costs one gate, and each is a perfect square, so it contributes a genuine *pair* of roots.

That second mechanism generalizes.

Proposition 8.2 (square-tower family; MV for $m \le 4$)**.** *Let $t_1 = 2$ and $t_{i+1} = t_i^2$, and set*

$$p_m(x) \;=\; x^2\,(x^2-1)\prod_{i=2}^{m+1}\bigl(x^2-t_i\bigr).$$

Then p_m has $z = 2m+3$ distinct integer roots and is computed by $\tau = 3m+4$ gates for $m \ge 1$ (3 gates when $m = 0$): three gates per two roots.

The family is a useful calibration precisely because it is *not* uniformly optimal. At $m = 0$ it gives 3 roots in 3 gates and the census says the minimum for 3 roots is exactly 3; at $m = 1$ it gives 5 roots in 7 gates where the census achieves 5 roots in 6. It therefore supplies upper bounds only, and it is the exhaustive scan below, not the construction, that decides optimality at seven.

8.2 Two arithmetics of opposite parity

Witnesses (A) and (B) exploit different mechanisms, and the mechanisms differ in parity. Folding by $q = x^2 - x$ identifies x with $1 - x$, so every value with integer preimages contributes exactly two roots: folding natively produces *even* root counts, and an odd count costs an extra linear factor. Squaring by $y = x^2$ has one self-paired value, 0, whose preimage is the single point $x = 0$: squaring natively produces *odd* counts, $2m + 3$.

Seven is odd, so the tower is native there and the interval needs its patch, and the two come out equal at ten gates. Across the exhaustive range an interval is always *among* the minimal-gate record sets, though it is not the only one: at $\tau = 5$ the four-root records include the interval $\{-1, 0, 1, 2\}$ and also the difference-of-squares set $\{\pm1, \pm2\}$, computed by $x^2 - (x^2 - 2)^2$, which is not an interval [MV]. The census therefore ends exactly where the two mechanisms tie, and never separates them.

Height does separate them. Witness (A) has height 56 against (B)'s 84, and at six roots the interval record has height 15 against 84 for the corresponding tower. This matches a direct measurement over the census witnesses: the minimal height of a polynomial in our catalog with z distinct integer roots is $1, 1, 2, 4, 15$ for $z = 2, \ldots, 6$ [MV], attained by intervals. Additively structured root sets are the height-cheap ones; multiplicatively structured ones buy their constants at doubly exponential magnitude. We record this because it constrains proof strategies: a lower-bound argument tuned to spread, multiplicatively built root sets will be blind to interval sets, and conversely.

8.3 The ninth gate, multiplicative half

By the last-gate lemma a 9-gate program is one gate over a depth-8 state, and since $z_{\max}(8) = 6$ the final gate of any 9-gate seven-rooter must involve the eighth value. When that gate is a multiplication, $f = v \cdot b$ and $z(f) = |R_v \cup R_b|$, so the question reduces to unions of root sets already known from the frontier, with no polynomial arithmetic in the inner loop.

Theorem 8.3 (MV)**.** *No 9-gate constant-free program whose final gate is a multiplication computes a polynomial with 7 distinct integer roots. Over all 1,048,460,912 depth-7 states, every one-gate extension and every operand, the maximum union size is exactly 6.*

The scan was gated before production: the same machinery run at threshold 6 returns 793 unions on a single partition, so it detects unions when they exist. This half is unconditional; root-set unions require no evaluation window.

8.4 The additive half, and what "windowed" costs

An additive final gate admits no such reduction, since root sets do not compose across addition. The instrument works with values instead: every operand and extension is evaluated on the window $[-32, 32]$ modulo a 31-bit prime, and $f = v \pm b$ vanishes at a point only if the residues agree, so counting modular agreements never misses a witness; candidates reaching seven agreements are promoted to exact construction and exact root counting. Two traps were fixed before production. First, 61-bit primes overflow `int64` under modular multiplication, so the modulus is 31-bit. Second, identically-zero polynomials agree at every window point and would otherwise flood the promotion stage. They are excluded, but only under a degree side-condition: agreement at all 65 window points is discarded when the candidate's degree bound is below 65, since a nonzero polynomial of degree below 65 cannot vanish at 65 points, and is *retained* when the degree bound is 65 or more. The side-condition matters, because a 9-gate polynomial can have degree up to 2^8, and discarding full agreement unconditionally would have thrown away exactly the high-degree witnesses the scan is looking for.

Emptiness established this way is *windowed*: on its own it excludes only witnesses all of whose roots lie in $[-32, 32]$, and a 9-gate program can build constants as large as 2^{256} by repeated squaring. The following turns that caveat from an unknown into a decidable criterion.

Lemma 8.4 (confinement; D, with the bound MV)**.** *Let $f \in \mathbb{Z}[x]$ be nonzero with distinct integer roots $r_1, \ldots, r_k$. Then $\prod_i (n - r_i)$ divides $f(n)$ for every $n \in \mathbb{Z}$. Consequently, if f has 7 distinct integer roots and c denotes its trailing (first nonzero) coefficient, then $|c| \leq 395$ forces every integer root of f to lie in $[-32, 32]$.*

Proof. The r_i are distinct integers, so $\prod_i (x - r_i)$ is monic in $\mathbb{Z}[x]$ and divides f in $\mathbb{Q}[x]$; by Gauss the cofactor lies in $\mathbb{Z}[x]$, and evaluating at n gives the divisibility. For the bound, suppose some root satisfies $|r_j| \geq 33$. If 0 is not a root then $c = f(0)$ and $|c| \geq \prod_i |r_i|$; the six remaining roots are distinct and nonzero, so their product has absolute value at least $|(-1)(1)(-2)(2)(-3)(3)| = 36$, giving $|c| \geq 33 \cdot 36 = 1188$. If 0 is a root, write $f = x^m g$ with $c = g(0)$; the six other roots are distinct and nonzero, and the five non-escaping ones give a product of absolute value at least 12, so $|c| \geq 33 \cdot 12 = 396$. Either way $|c| \geq 396$, contradicting $|c| \leq 395$. □

The constant is sharp: searching seven-root sets containing an escaping root returns minimum trailing coefficient exactly 396, at $\{-3, -2, -1, 0, 1, 2, 33\}$ [MV]. We note that the two witnesses of Proposition 8.1 have trailing coefficients 48 and -64, an order of magnitude inside the criterion: the shapes that actually reach seven roots cheaply are exactly the ones the window sees.

Lemma 8.4 does not make the additive scan unconditional. It prices the gap, and we measured the price. Constructing every candidate $f = t \pm b$ exactly on a sample of 6,400,000 candidates drawn from four hash-partitioned frontier files, and reading each one's true trailing coefficient [MV]:

identically zero (excluded)	93,898	1.467%
confined by Lemma 8.4	6,301,361	99.9248%
excluded by $\deg f < 7$	4,619	0.0732%
window-limited, $\deg f \geq 7$	122	0.0019%

the last three percentages being of the 6,306,102 nonzero candidates. The degree row is arithmetic rather than a lemma: seven distinct roots require degree at least seven, so those candidates are excluded with no reference to the window at all. Together, 99.998% of nonzero candidates are decided independently of the evaluation window, and the residual is a structurally narrow class (degrees 8, 9 and 12; 64 distinct trailing coefficients between 512 and 38,220).

We state the consequence in the form the evidence supports, and no stronger: emptiness on the additive side holds outright for 99.998% of the search space, and is windowed only on a measured residual near 0.002%. These are sample estimates from four of 256 partitions, not exhaustive counts. Making the residual exhaustive is affordable precisely because it is rare, and is the natural next step.

9 The three worlds, measured

It is folklore that the conjecture's analogue fails over finite fields, and that $x^{p-1}-1$ is the reason; this is why the statement is always made in characteristic zero. We record here that the finite-field question has an exact answer, obtained on the *same* enumerated programs as the integer census, so that the two ladders are comparable term by term.

For a prime p write $z_p(f)$ for the number of distinct roots of f in $\mathbb{F}_p$, defined only when $f \not\equiv 0 \pmod p$, and set

$$z^p_{\max}(\tau) \;=\; \max\bigl\{\, z_p(f) \;:\; \tau(f) \le \tau,\ p \text{ prime},\ f \not\equiv 0 \pmod p \,\bigr\}.$$

Proposition 9.1 (D, with both halves MV). $z^p_{\max}(\tau) = 2^{\tau-1}$ *for every* $\tau \ge 1$.

Proof. Upper bound. For a value v of a constant-free program let $\mu(v)$ count the *distinct* multiplicative gates in its sub-DAG. We claim $\deg v \le 2^{\mu(v)}$, by induction. The input has degree 1 with $\mu = 0$, and constants have degree 0. For an additive gate $v = a \pm b$, the sub-DAG of v contains those of a and b, so $\mu(v) \ge \max(\mu(a), \mu(b))$ and $\deg v \le \max(\deg a, \deg b) \le 2^{\max(\mu(a),\mu(b))} \le 2^{\mu(v)}$. For a multiplicative gate $v = ab$ we have $\mu(v) = |M(a) \cup M(b)| + 1 \ge \max(\mu(a), \mu(b)) + 1$, where $M(\cdot)$ denotes the set of multiplicative gates in a sub-DAG, and hence

$$\deg v = \deg a + \deg b \le 2^{\mu(a)} + 2^{\mu(b)} \le 2^{\max(\mu(a),\mu(b))+1} \le 2^{\mu(v)}.$$

Counting *distinct* gates, so that shared subexpressions are not charged twice, is what makes the multiplicative step close: bounding the exponent by $\mu(a) + \mu(b) + 1$ instead would exceed $\mu(v)$ whenever a and b share gates.

Now a sub-DAG with no additive gate computes only $\pm x^k$ or 0, since the free constants are $-1, 0, 1$ and products of those remain in $\{-1, 0, 1\}$; such a polynomial has at most one distinct root in any field. So if f is not of that form its sub-DAG contains an additive gate, whence $\mu(f) \le \tau(f) - 1$ and $\deg f \le 2^{\tau-1}$. Finally a polynomial that is nonzero over a field has at most its degree many roots, and reduction mod p can only lower the degree.

Lower bound. By Dirichlet there is a prime $p \equiv 1 \pmod{2^k}$ for every k. Modulo such p the polynomial $x^{2^k} - 1$ has exactly $\gcd(2^k, p-1) = 2^k$ roots, and it is computed by k squarings and one subtraction, so $\tau \le k+1$. Taking $k = \tau - 1$ gives $2^{\tau-1}$. □

Both halves were checked mechanically: the degree bound holds on every non-monomial in the complete census to depth 6, where it is moreover attained exactly ($\max\deg = 2^{\tau-1}$ at each τ, against 2^τ for monomials), and the construction was verified for $k = 1, \ldots, 8$ at $p = 3, 5, 17, 17, 97, 193, 257, 257$ with root counts $2, 4, 8, 16, 32, 64, 128, 256$ [MV]. We note that the Fermat primes, which the small cases suggest, are not needed: only $p \equiv 1 \pmod{2^k}$ is.

9.1 The real ladder

The real analogue of the conjecture is known to fail. We measured how far, on the same enumerated programs, counting distinct real roots exactly: take the square-free part $f/\gcd(f, f')$ and count real roots by Sturm's theorem in exact rational arithmetic, with no floating point at any stage. The instrument was gated on seven known answers first, among them x^2+1 (none), x^2 (a double root counted once), T_4 (four), and $(x^2-1)(x^2-4)(x^2-16)$ (six). Writing $z^{\mathbb{R}}_{\max}(\tau)$ for the maximum over τ-gate polynomials:

τ	1	2	3	4	5	6	7	8
$z_{\max}(\tau)$	1	2	3	3	4	5	5	6
$z^{\mathbb{R}}_{\max}(\tau)$	1	2	3	4	6	8	–	–
$z^{p}_{\max}(\tau)$	1	2	4	8	16	32	64	128

(The real row is exhaustive through $\tau = 6$. That row was obtained without a depth-6 census: the last-gate lemma applied to the 778,087-state depth-5 frontier enumerates all 134,497 new $\tau = 6$ polynomials in 180 s, and the run carried a known-answer gate on the same set, requiring the *integer* maximum to come out at 5 before the real number was reported. Reaching $\tau = 7$ needs the 25.8M-state depth-6 frontier resident, which we defer.) The three worlds separate at three different points: $\mathbb{F}_p$ leaves $\mathbb{Z}$ at $\tau = 3$ and saturates the degree ceiling $2^{\tau-1}$ at every τ; $\mathbb{R}$ leaves $\mathbb{Z}$ at $\tau = 4$ and does *not* saturate it (4 against 8, then 6 against 16); $\mathbb{Z}$ trails both.

The real record at $\tau = 5$ is worth writing out, because it is produced by the same difference-of-squares mechanism that Section 5 identifies as the integer record-maker:

$$a = x \cdot x, \quad b = a - 1, \quad c = b \cdot b, \quad d = c - a, \quad e = b \cdot d,$$

$$e \;=\; (x-1)(x+1)(x^2-x-1)(x^2+x-1).$$

Five gates, *six* distinct real roots $\pm 1, \pm\varphi, \pm\varphi^{-1}$ with φ the golden ratio, and exactly *two* distinct integer roots [MV]. The program is cheap and the roots are there; four of them are simply irrational. This is the conjecture's content in one line: what limits $\mathbb{Z}$ is arithmetic, not the cost of manufacturing roots. It also shows the real ladder is not merely the Chebyshev ladder, which buys a doubling per three gates ($T_2 = 2x^2 - 1$ costs three, and composition repeats it) and would give only four roots here.

The $\tau = 6$ record exposes the mechanism. It is

$$x^8 - 7x^6 + 14x^4 - 7x^2 + 1 \;=\; (x^4 - 3x^2 + 1)(x^4 - 4x^2 + 1),$$

whose left factor is exactly the $\tau = 4$ record $g_1 = (x^2-1)^2 - x^2$. So the step from four gates to six is $g \mapsto g \cdot (g - x^2)$, costing *two* gates because x^2 is already in the state, and *doubling* the real root count. It beats the obvious construction: a Chebyshev tower costs three gates per doubling ($T_2 = 2x^2 - 1$ is three gates and composition repeats it), so it reaches only about $2^{\tau/3}$. And it is not merely a good construction here, it matches the exhaustive census at both $\tau = 4$ and $\tau = 6$.

It also stalls, in the same manner as the integer-side towers of Section 6. Iterating the step gives distinct real roots against degree

gates	4	6	8	10	12
deg	4	8	16	32	64
real	4	8	16	28	48

[MV]: three clean doublings, reaching 16 distinct real roots at 8 gates, after which the iterate ceases to be totally real and the successive ratios fall to 1.75 and 1.71.

We claim no rate for $z^{\mathbb{R}}_{\max}$: six points cannot distinguish $\Theta(2^{\tau/2})$ from anything nearby, the family's own rate decays past $\tau = 8$, and the paper needs only the known failure of the real analogue.

One structural point is worth recording. The $\tau = 6$ real record has *zero* integer roots, while the $\tau = 6$ integer record has five: at equal cost the two worlds are optimized by entirely disjoint polynomials. The real ladder is not the integer ladder with further roots attached, but a different extremal problem that happens to share the difference-of-squares mechanism at the bottom.

Setting the three ladders side by side is the point of this section, and the $\mathbb{F}_p$ row is the one that is now exact. The ladders agree at $\tau \leq 2$ and separate from $\tau = 3$ onward. Over $\mathbb{F}_p$ the count doubles with every gate, which is the fastest the degree bound permits, and the extremal programs are the cheapest available: repeated squaring, then subtract one. The same polynomial $x^{2^k} - 1$ has exactly two integer roots, ± 1, for every k.

The mechanism is the one the three-worlds reading predicts. In $\mathbb{F}_p^{\times}$, cyclic of order $p-1$, the equation $x^d = 1$ has $\gcd(d, p-1)$ solutions, up to d of them; in $\mathbb{Z}^{\times} = \{\pm 1\}$ it has at most two, for every d. The exponential ladder is precisely the 2^k-torsion that the cyclic group carries and the integers do not. In this light the conjecture asserts that no program shape recovers over $\mathbb{Z}$ what the cyclotomic one achieves over $\mathbb{F}_p$, and the census is consistent with that: through $\tau = 8$ nothing over $\mathbb{Z}$ does better than linear growth. What Proposition 9.1 contributes is that the comparison is now exact on one side, so the gap being conjectured is a gap between a proved exponential and a measured linear rate rather than between two informal impressions.

10 Limits, and what would count as progress

Nothing above bears on the conjecture asymptotically: censuses to depth 7 cannot distinguish polynomial from superpolynomial growth, and our theorems close construction classes rather than open the bounding problem. The honest state of the measured landscape is: every mechanism observed or excluded so far pays $\Theta(1)$ gates per root, the growth function's first structure (plateaus at 4 and 7) is the visible cost of constant-building, and the geometric-progression family remains the best known factory, linear-rate, exactly as when Rojas noted that the conjecture fails for $\kappa < 1$ [8].

Concrete open items this record makes precise: (i) the $\{9, 10\}$ window for the seven-root threshold, and $z_{\max}(9)$ (a depth-8 frontier build would need $\sim$1 TB of scratch: a construction hunt with the corrected cost model is the nearer route); (ii) extend the integer frontier past Markström's length 11 [7], including his monotonicity question for $\tau(n!)$; (iii) measure the growth of the valuation-spectrum bound $s \leq N_p(s) \leq s(s+1)/2$ of [8], where the true rate is stated as open; (iv) the census's 2-adic instrumentation records that all depth-5–7 record polynomials concentrate their roots on the valuations $\{0, 1\}$ [MV], consistent with the pressure sitting on the digit-conjecture side of Rojas' reduction [C].

Data and code availability

All code, hypotheses, run artifacts, verdicts and the wiki are committed at `github.com/fsantibanezleal/CAOS_R` under `problems/computation-complexity/tau-conjecture/` (experiments `EXP-001`–`EXP-005`; package `tclib` with its test suite). All computations are exact integer arithmetic in Python 3.13

standard library; total compute for every result in this paper is under twenty hours on one desktop machine (24 cores; the depth-8 pipeline used 20 processes and ∼29 GB of scratch disk; peak RAM ∼12 GB). The constructed depth-7 frontier (1,048,460,912 states, 28 GB) and the polynomial catalog are retained as reusable data assets with hashes recorded in the repository.

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